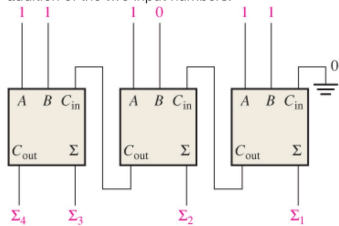
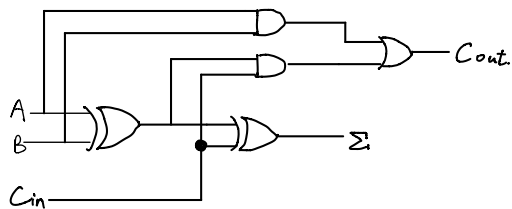


4. For the parallel adder in Figure 6-69, determine the complete sum by analysis of the logical operation of the circuit. Verify your result by longhand addition of the two input numbers.



4.



first box: $A=1, B=1, C_{in}=0$

$$\Sigma_1 = (A \oplus B) \oplus C = 0 \quad \text{Cout} = AB + C_{in}(A \oplus B) = 1$$

second box: $A=1, B=0, C_{in}=1$

$$\Sigma_2 = (1 \oplus 0) \oplus 1 = 0 \quad \text{Cout} = 1 \cdot 0 + 1 \cdot (1 \oplus 0) = 1$$

Third box: $A=1, B=1, C_{in}=1$

$$\Sigma_3 = (1 \oplus 1) \oplus 1 = 1 \quad \Sigma_4 = \text{Cout} = 1 \cdot 1 + 1 \cdot (1 \oplus 1) = 1$$

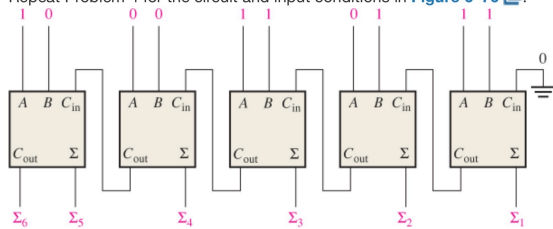
$$\Sigma_4 \Sigma_3 \Sigma_2 \Sigma_1 = 1100$$

where $A+B = 111$

$$\begin{array}{r} 111 \\ + 101 \\ \hline 1100 \end{array}$$

verified.

5. Repeat Problem 4 for the circuit and input conditions in Figure 6-70.



5. $A = 10101$
 $B = 00111$

first box: $A = 1$ $B = 1$ $C_{in} = 0$
 $\Sigma_1 = (1 \oplus 1) \oplus 0 = 0$ $C_{out} = 1 + 1 \cdot 0 = 1$

second box: $A = 0$ $B = 1$ $C_{in} = 1$
 $\Sigma_2 = (0 \oplus 1) \oplus 1 = 0$ $C_{out} = 0 + 1 \cdot 1 = 1$

third box: $A = 1$ $B = 1$ $C_{in} = 1$
 $\Sigma_3 = (1 \oplus 1) \oplus 1 = 1$ $C_{out} = 1$

fourth box: $A = 0$ $B = 0$ $C_{in} = 1$
 $\Sigma_4 = 1$ $C_{out} = 0 + 1 \cdot 0 = 0$

fifth box: $A = 1$ $B = 0$ $C_{in} = 0$
 $\Sigma_5 = 1$ $C_{out} = \Sigma_6 = 0$

$$\Sigma_{654321} = 011100$$

$$\begin{array}{r} A+B = 10101 \\ + 00111 \\ \hline 11100 \end{array}$$

verified.

7.

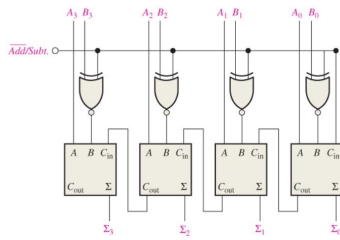


Figure 6-71

7. For the circuit in **Figure 6-71**, assume the inputs are

$$\overline{\text{Add/Subt.}} = 1$$

, $A = 1001$ and $B = 1100$. What is the output?

$$A = 1001$$

$$B = 1100$$

first box: $A_0 = 1$ $B_0 = 0$ $C_{in} = 1$

$$A = 1 \quad B = \overline{0 \oplus 1} = 0$$

$$\Sigma_0 = (1 \oplus 0) \oplus 1 = 0 \quad \text{Count} = 0 + 1(1 \oplus 0) = 1$$

second box: $A_1 = 0$ $B_1 = 0$ $C_{in} = 1$

$$A = 0 \quad B = \overline{0 \oplus 1} = 0$$

$$\Sigma_1 = 1 \quad \text{Count} = 0 + 0 = 0$$

3rd box: $A_2 = 0$ $B_2 = 1$ $C_{in} = 0$

$$A = 0 \quad B = \overline{1 \oplus 1} = 0$$

$$\Sigma_2 = 1 \quad \text{Count} = 0 + 0 = 0$$

4th box: $A_3 = 1$ $B_3 = 1$ $C_{in} = 0$

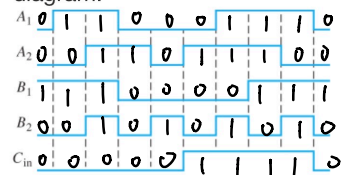
$$A = 1 \quad B = \overline{1 \oplus 1} = 0$$

$$\Sigma_3 = 0 \quad \text{Count} = 1$$

$$\Sigma_{3210} = 0110$$

8.

8. The input waveforms in **Figure 6-72** are applied to a 2-bit adder. Determine the waveforms for the sum and the output carry in relation to the inputs by constructing a timing diagram.

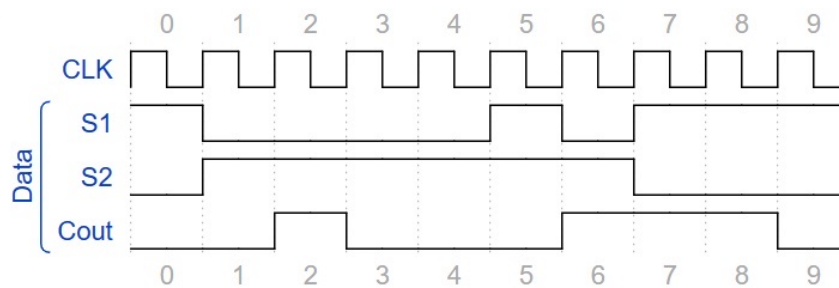


$$\Sigma_1 = S_1 = (A_1 \oplus B_1) \oplus C_{in} = 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 1$$

$$C_{out1} = A_1 B_1 + C_{in} (A_1 \oplus B_1) = 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 0$$

$$\Sigma_2 = S_2 = (A_2 \oplus B_2) \oplus C_{out1} = 0 \ 1 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0$$

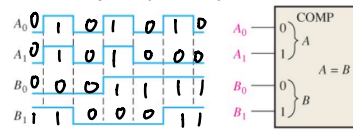
$$C_{out2} = \Sigma_3 = (A_2 B_2) + C_{out1} (A_2 \oplus B_2) = 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 0$$



13. The waveforms in **Figure 6-73**

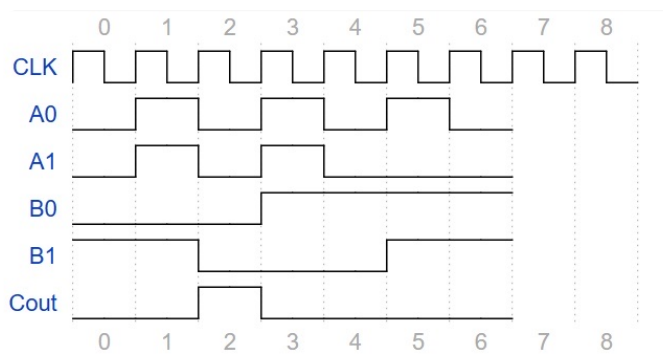
13.

are applied to the comparator as shown. Determine the output ($A = B$) waveform.



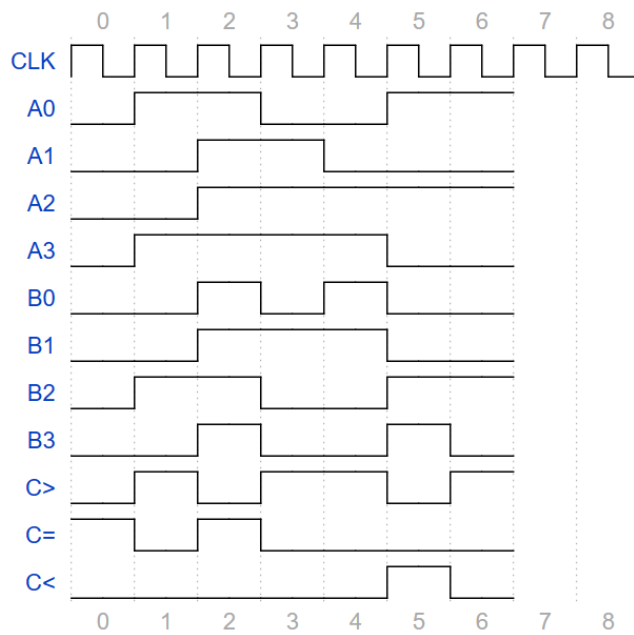
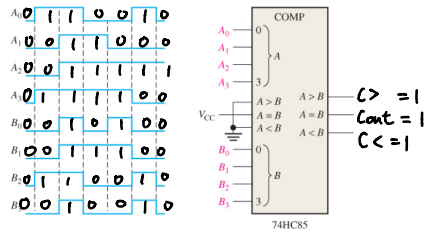
0 0 1 0 0 0 0

$C_{out} = 1$ when $A_0 = B_0$ and $A_1 = B_1$



14.

14. For the 4-bit comparator in **Figure 6-74**, plot each output waveform for the inputs shown. The outputs are active-HIGH.



19.

$$ABCD = | | | |$$

$$X = 0001, 1100, 1011, 1010$$

AB \ CD	00	01	11	10
00			X	
01	X			
11				(X)
10				(X)

$$X = \bar{A}\bar{B}\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}C$$

